SAECOIN — OPEN TECHNICAL DECISIONS

Document: Open_Decisions.md
Project: SAECOIN
Ticker: SAC (Provisional)
Version: 1.0
Status: Open — Decisions Pending
Purpose: Track unresolved technical, economic, security and operational decisions before implementation.

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1. PURPOSE

This document records all important decisions that have not yet been finalized for SAECOIN.

The purpose is to:

- Prevent inconsistent implementation.
- Identify missing protocol specifications.
- Track technical risks.
- Record approved decisions.
- Avoid changing consensus rules unexpectedly after implementation.
- Prepare the project for independent technical review.

No unresolved consensus-critical decision should be silently assumed by developers.

2. DECISION STATUS

Use these statuses:

- OPEN: No final decision.
- RESEARCH: Technical investigation required.
- REVIEW: Proposed solution awaiting review.
- APPROVED: Formally approved.
- REJECTED: Proposal rejected.
- DEFERRED: Intentionally postponed.

Each decision must have an owner, supporting evidence and approval date.

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3. BLOCKCHAIN CONSENSUS DECISIONS

DEC-001: Proof of Work Algorithm

Status: OPEN

Questions:

- Which PoW algorithm will SAECOIN use?
- Should it be optimized for CPU mining?
- Should it support GPU mining efficiently?
- What hardware resistance properties are required?
- What are the verification performance requirements?

Importance: CRITICAL

Required Before: Consensus implementation.

DEC-002: ASIC Resistance Strategy

Status: OPEN

Questions:

- Should the algorithm be memory-hard?
- Should memory requirements increase over time?
- How will GPU and ASIC performance be evaluated?
- What evidence will be required before describing SAECOIN as ASIC-resistant?

Importance: HIGH

Required Before: Mining algorithm finalization.

Note: ASIC resistance cannot be guaranteed.

DEC-003: Block Hash Algorithm

Status: OPEN

Questions:

- Which cryptographic hash function will be used?
- Will the block header use one hash or multiple hashing stages?
- How will the hash algorithm be domain-separated?
- How will future algorithm changes be handled?

Importance: CRITICAL

Required Before: Block header implementation.

DEC-004: Difficulty Adjustment Algorithm

Status: OPEN

Questions:

- Which difficulty adjustment formula will be used?
- How frequently will difficulty change?
- How will sudden hash-rate changes be handled?
- How will timestamp manipulation be mitigated?
- What prevents difficulty manipulation through short-term mining attacks?

Importance: CRITICAL

Required Before: Consensus implementation.

DEC-005: Block Timestamp Rules

Status: OPEN

Questions:

- What is the maximum permitted future timestamp?
- Will median-time-past be used?
- How many previous blocks will be considered?
- What timestamp rules apply during reorganizations?

Importance: HIGH

DEC-006: Chain Selection Rules

Status: OPEN

Questions:

- Will the network select the valid chain with the most cumulative work?
- How will equal-work competing chains be handled?
- How will nodes recover from deep reorganizations?
- What are the maximum operational reorganization warning thresholds?

Importance: CRITICAL

DEC-007: Block Size or Weight

Status: OPEN

Questions:

- What is the maximum block size or weight?
- Will transaction weight be used?
- How will oversized blocks be rejected?
- What are the expected storage and bandwidth requirements?

Importance: HIGH

DEC-008: Block Time

Status: REVIEW

Current Proposal: 10 minutes.

Questions:

- Is 10 minutes appropriate for SAECOIN?
- What confirmation experience is expected?
- How does block time affect network security and transaction throughput?

Importance: HIGH

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4. MONETARY POLICY DECISIONS

DEC-009: Maximum Supply

Status: REVIEW

Current Proposal: 21,000,000 SAC.

Questions:

- Will 21 million SAC remain the maximum supply?
- How will the maximum supply be enforced?
- How will integer rounding affect total issuance?

Importance: CRITICAL

DEC-010: Initial Block Reward

Status: REVIEW

Current Proposal: 50 SAC.

Questions:

- Is the initial reward appropriate?
- How will reward validation be implemented?
- What happens if the reward calculation exceeds the permitted supply?

Importance: CRITICAL

DEC-011: Halving Interval

Status: REVIEW

Current Proposal: 210,000 blocks.

Questions:

- Will the reward halve every 210,000 blocks?
- How will rounding be handled?
- What happens when the subsidy becomes zero?
- How will the complete issuance schedule be verified?

Importance: CRITICAL

DEC-012: Transaction Fees

Status: OPEN

Questions:

- Will users specify fees manually?
- Will wallets estimate fees automatically?
- Will minimum relay fees exist?
- How will fee calculation prevent spam?
- Will miners receive all transaction fees?

Importance: HIGH

DEC-013: Coinbase Maturity

Status: OPEN

Questions:

- How many confirmations are required before mining rewards can be spent?
- Will the maturity period apply to all coinbase outputs?
- How will reorganizations affect immature rewards?

Importance: HIGH

DEC-014: Dust and Minimum Output

Status: OPEN

Questions:

- What is the minimum permitted transaction output?
- How will dust outputs be identified?
- Will dust thresholds depend on transaction size and relay policy?

Importance: MEDIUM

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5. TRANSACTION AND CRYPTOGRAPHY DECISIONS

DEC-015: Digital Signature Algorithm

Status: OPEN

Questions:

- Which signature algorithm will be used?
- What are its security assumptions?
- Is the implementation suitable for Rust and Android?
- How will signature verification be standardized?

Importance: CRITICAL

DEC-016: Public Key Format

Status: OPEN

Questions:

- Which public key encoding will be used?
- Will compressed public keys be supported?
- How will key serialization be versioned?

Importance: HIGH

DEC-017: Address Format

Status: OPEN

Questions:

- What address encoding will SAECOIN use?
- What will the mainnet prefix be?
- What will the testnet prefix be?
- How will address checksums work?
- How will invalid addresses be detected?

Importance: HIGH

DEC-018: Transaction Identifier

Status: OPEN

Questions:

- Which hash will identify a transaction?
- Will witness or signature data affect the transaction ID?
- How will transaction malleability be handled?

Importance: HIGH

DEC-019: Transaction Serialization

Status: OPEN

Questions:

- What exact byte format will transactions use?
- Will serialization be canonical?
- How will malformed or non-canonical transactions be rejected?
- How will transaction format upgrades work?

Importance: CRITICAL

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6. NETWORK DECISIONS

DEC-020: P2P Port

Status: OPEN

Questions:

- Which default port will mainnet use?
- Which port will testnet use?
- How will port conflicts be handled?

Importance: MEDIUM

DEC-021: Network Magic Bytes

Status: OPEN

Questions:

- What unique magic bytes will identify SAECOIN messages?
- How will messages from other networks be rejected?

Importance: HIGH

DEC-022: Peer Discovery

Status: OPEN

Questions:

- Will DNS seeds be used?
- Will static bootstrap peers be included?
- Will peer addresses be stored locally?
- How will malicious peer discovery be mitigated?

Importance: HIGH

DEC-023: Peer Limits

Status: OPEN

Questions:

- What is the maximum number of inbound peers?
- What is the maximum number of outbound peers?
- What connection limits are appropriate for ordinary computers?

Importance: MEDIUM

DEC-024: Network Upgrade Mechanism

Status: OPEN

Questions:

- How will protocol upgrades be activated?
- Will upgrades require a new software release?
- How will incompatible consensus changes be coordinated?
- How will users be informed of mandatory upgrades?

Importance: CRITICAL

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7. WALLET DECISIONS

DEC-025: Wallet Recovery Standard

Status: OPEN

Questions:

- Will SAECOIN use a standardized mnemonic recovery format?
- How many recovery words will be used?
- How will recovery compatibility be maintained?
- Will users be able to import existing private keys?

Importance: HIGH

DEC-026: Wallet Encryption

Status: OPEN

Questions:

- Which key derivation function will be used?
- How will wallet passwords be protected?
- Which authenticated encryption method will be used?
- How will password changes work?

Importance: HIGH

DEC-027: Wallet Architecture

Status: REVIEW

Current proposal:

- Android: Kotlin + Jetpack Compose.
- Windows: Tauri + Rust + TypeScript.

Questions:

- Will both wallets use the same Rust wallet core?
- How will wallet-core compatibility be maintained?
- Will hardware wallet support be considered later?

Importance: HIGH

DEC-028: Wallet Synchronization

Status: OPEN

Questions:

- Will wallets connect directly to full nodes?
- Will a lightweight synchronization method be supported?
- How will wallets verify blockchain information?
- What privacy trade-offs will be accepted?

Importance: HIGH

DEC-029: Confirmation Display

Status: REVIEW

Current suggestion: Six confirmations for higher-confidence payment handling.

Questions:

- Will six confirmations be the default display policy?
- Should merchants choose their own confirmation thresholds?
- How will the wallet communicate that confirmations reduce risk but do not create absolute finality?

Importance: MEDIUM

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8. MINING DECISIONS

DEC-030: Initial Mining Model

Status: REVIEW

Current proposal: Solo mining first.

Questions:

- Will solo mining be the only initial supported model?
- When should mining pool support be introduced?
- What software interface will future pools require?

Importance: MEDIUM

DEC-031: Mining Software Architecture

Status: OPEN

Questions:

- Will CPU and GPU mining share a common interface?
- Will GPU mining use a separate mining engine?

- How will mining jobs be communicated to devices?
- How will stale work be handled?

Importance: HIGH

DEC-032: Mining Performance Targets

Status: OPEN

Questions:

- What CPU performance should be measured?
- What GPU performance should be measured?
- What memory usage is acceptable?
- What hardware will be included in benchmark testing?

Importance: MEDIUM

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9. DATABASE AND NODE DECISIONS

DEC-033: Node Database

Status: REVIEW

Current proposal: RocksDB.

Questions:

- Is RocksDB appropriate for the initial implementation?
- What are its Windows compatibility requirements?
- How will database corruption be handled?
- What backup and recovery mechanisms are required?

Importance: HIGH

DEC-034: Pruning

Status: OPEN

Questions:

- Will nodes initially store the complete blockchain?
- Will pruned-node support be introduced?
- How will pruned nodes synchronize and validate new blocks?
- What historical data must remain available?

Importance: MEDIUM

DEC-035: Node Resource Requirements

Status: OPEN

Questions:

- What minimum RAM is required?
- What disk capacity is recommended?
- What CPU requirements apply?
- What bandwidth is expected?
- What hardware is required for long-term operation?

Importance: MEDIUM

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10. DEVELOPMENT AND RELEASE DECISIONS

DEC-036: Rust Workspace Architecture

Status: REVIEW

Current proposal: Rust-based monorepo.

Questions:

- Which components will be independent Rust crates?
- Which components will use separate technology stacks?
- How will Android, Windows and explorer projects share protocol definitions?
- How will build dependencies be managed?

Importance: HIGH

DEC-037: Continuous Integration

Status: OPEN

Questions:

- Which CI platform will be used?
- Will Windows and Linux builds be tested?
- Will automated security checks run on every change?
- How will release builds be verified?

Importance: MEDIUM

DEC-038: Release Signing

Status: OPEN

Questions:

- Which signing mechanism will be used?
- Who controls official release signing keys?
- How will signing keys be protected?
- How will users verify official releases?

Importance: HIGH

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11. MAINNET AND GOVERNANCE DECISIONS

DEC-039: Mainnet Launch Requirements

Status: REVIEW

Current reference: Mainnet_Launch_Checklist.md.

Questions:

- What minimum public testnet duration is required?
- What independent security reviews are required?
- What conditions constitute launch approval?

Importance: CRITICAL

DEC-040: Project Governance

Status: OPEN

Questions:

- Who approves protocol changes?
- How will contributors propose changes?
- How will consensus-critical changes be reviewed?
- How will emergency fixes be coordinated?
- How will the community be informed?

Importance: HIGH

DEC-041: Project Treasury

Status: OPEN

Questions:

- Will SAECOIN have a project treasury?
- If yes, how will it be funded?
- Who controls treasury funds?
- Will treasury rules be enforced by consensus or external governance?

Importance: HIGH

Important: A treasury or developer allocation must not be introduced accidentally through coinbase rules or undocumented issuance.

DEC-042: Exchange and Commercial Strategy

Status: DEFERRED

Questions:

- Will SAECOIN pursue exchange listings?
- What legal and technical requirements apply?
- How will users be protected from misleading investment claims?
- What public disclosures will be required?

Importance: MEDIUM

This decision does not need to block initial protocol research, but legal review is required before relevant commercial activities.

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12. DECISION PRIORITY

Priority| Decisions| Requirement
Critical| DEC-001, 003, 004, 006, 009, 010, 011, 015, 019, 024, 039| Resolve before production consensus implementation or mainnet
High| DEC-002, 005, 007, 012, 013, 016, 017, 018, 022, 026, 027, 028, 031, 033, 036, 038, 040, 041| Resolve before relevant component is finalized
Medium| DEC-008, 014, 020, 023, 025, 029, 030, 032, 034, 035, 037, 042| Resolve according to implementation and launch dependencies

The priority table is a planning aid. A decision may be upgraded if technical review identifies additional risk.

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13. DECISION RECORD TEMPLATE

Use this template whenever a decision is made.

Decision ID

DEC-XXX

Title

Decision name.

Status

OPEN / RESEARCH / REVIEW / APPROVED / REJECTED / DEFERRED

Problem

Describe the problem that requires a decision.

Options Considered

1. Option A
2. Option B
3. Option C

Selected Option

Final selected option.

Technical Reasoning

Explain why the option was selected.

Risks

Document technical and operational risks.

Evidence

Link benchmarks, research, test results or independent reviews.

Approval

Approved by:

Approval date:

Implementation Impact

List the documents, software components and tests that must be updated.

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14. FINAL RULES

1. Never implement an unresolved consensus-critical decision by guessing.
2. Never change monetary rules without a documented protocol review.
3. Never modify genesis parameters after mainnet launch.
4. Keep testnet and mainnet parameters separate.
5. Record all approved decisions.
6. Update related documentation whenever a decision changes.
7. Require independent review for consensus-critical changes.
8. Do not describe proposed parameters as finalized.
9. Do not promise ASIC resistance without evidence.
10. Do not begin mainnet launch preparation while critical decisions remain unresolved.

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15. CURRENT PROJECT STATE

At this documentation stage:

- SAECOIN remains a planned independent blockchain.
- Proof of Work is the selected consensus family.
- UTXO is the selected transaction model.
- CPU and GPU mining are project objectives.
- The proposed monetary parameters remain subject to verification.
- Consensus-critical algorithms remain undecided.
- Wallet cryptographic standards remain undecided.
- Mainnet genesis parameters remain undecided.
- No production blockchain implementation is implied by this document.

Next Milestone: Complete technical research, resolve critical decisions, obtain independent review, and update the relevant specifications before beginning consensus-critical implementation.